

Determination of Critical Parameters in Platelet Margination

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Abstract—An investigation of margination dependence on hematocrit, platelet shape, and viscosity ratio of plasma to cytoplasm is presented. Whole blood is modeled as a suspension of deformable red blood cells (RBCs) and rigid platelets in a viscous liquid. The fluid phase is simulated using the lattice-Boltzmann method, the RBC membranes are modeled with a coarse-grained spectrin-link method, and the dynamics of rigid particles are updated using Newton's equations of motion for axisymmetric shapes. The results emphasize that an increase in hematocrit increases the rate of margination. The viscosity ratio between the interior cytoplasm and suspending fluid can considerably alter the rate of margination. The aspect ratio of surrogate platelet particles influences the rate of margination as well. Spherical particles tend to migrate more quickly than disks. Highly viscous or rigid RBCs slow down margination.

Keywords—Lattice-Boltzmann, Spectrin link, Red blood cell.

ABBREVIATIONS

RBC	Red blood cell
LB	Lattice-Boltzmann
SL	Spectrin link

INTRODUCTION

Rapid thrombi formation, or hemostasis, in response to arterial injuries is essential for bleeding arrest to avoid excessive blood lost. In addition to hemostasis, platelet-rich thrombi can occlude atherosclerotic arteries. Such occlusive thrombi can cause ischemic cardiac arrest in less than an hour.^{16,17} Rapid thrombus growth rates

requires a high rate of transport of platelets to the wall. The mechanism of platelet margination may guide us in identifying risk factors associated with atherothrombosis, and ways to prevent arterial occlusions.

Since platelets are the predominant components of arterial thrombi,^{7,17} their transport rate to the thrombotic site plays an important role in the rate of thrombus growth. The presence of red blood cells (RBCs) in whole blood controls the rate of platelet flux to the vessel walls increasing it by several orders of magnitude compared to transport due to Brownian motion.⁶¹ This higher rate of transport and drift of platelets towards the vessels walls due to RBCs is termed enhanced diffusivity.^{27,51} Studies on thrombus growth employing continuum transport models, including phenomenological models of enhanced diffusivity, suggest that platelet transport may be a critical rate limiting process in thrombus growth at relevant pathophysiological shear rates.^{5,50}

In addition to platelet concentration at the vicinity of vessel wall, platelets dynamics at this region may affect the rate of adhesion of platelets to the site of damage by changing collision frequency rate, collision contact time, and available surface area for contact.^{29,38} Due to the small time and length scales and optical properties of blood, the platelets dynamics near the vessel wall are not well understood. Platelets motions may be analogous to rolling motion of leukocytes on inflamed endothelium, but may exhibit more of a flipping motion due to its discoid shape. The bonds between platelet receptor glycoproteins GPIb and immobilized von Willebrand factor may be important in mediating and controlling this rolling motion.³⁶ The flat shape of platelets, however, is unsuitable for rolling compared to the spherical shape of leukocytes. The study by Pozrikidis⁴⁴ demonstrated that the rotation rate of a platelet-shaped particle undergoing flipping motions is exaggerated when adhesive forces and torques are not modeled. Experiments by Maxwell *et al.*³⁵ have identified sliding as the most frequent mode of

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platelet movement near the surface wall increasing the surface area available for contact. More recent theoretical studies show that, in addition to binding kinetics, hemodynamic parameters such as hematocrit, the thickness of the RBC-depleted layer, and morphological properties of platelets are important factors influencing platelet aggregation rate.⁴⁸

In small vessels, absent of RBCs,² platelets form a dilute suspension of rigid particles and migrate to the Segré-Silberberg equilibrium position⁴⁶ of $r/R \approx 0.6$. When RBCs are included, platelets migrate radially outward toward the vessel wall² and ultimately reside in the RBC-free layer. Binary suspensions of vesicles of equal volume with different levels of rigidity have shown to demonstrate margination-like physics.³² According to the experiments of Eckstein *et al.*²³ deformable RBCs are required for margination to occur. In this study, we focus on the process of margination in the presence of deformable RBCs. Therefore, the process of margination is reliant on differences in shape and/or rigidity of blood constituents which can be studied by modeling blood as a binary suspension of deformable RBCs and rigid platelets.

Three important factors in the platelet margination process considered in this study are hematocrit, viscosity ratio between the suspending fluid and cytoplasm, and platelet aspect ratio. The results presented in this work emphasize already known important hemodynamic parameter of hematocrit while revealing that viscosity ratio and platelet shape are also important parameters in the margination process. Shear rate^{1,47} and RBC deformability are also important parameters, but are not investigated in this work. The RBC-depleted layer thickness and platelet shape influence platelet dynamics at the vicinity of the wall. Due to the particulate nature of enhanced diffusivity and margination processes, numerical study of these phenomena requires a computational model that includes RBC and platelet mechanics and particle-particle and particle-fluid interactions in blood flow. This work is based on the hypothesis that in addition to hematocrit, the platelet margination rate depends also on platelet morphology and the viscosity ratio between the cytoplasm within the RBC and the suspending fluid.^{23,47,58}

Simulations of Cellular Blood Flow

The dynamic behavior of RBCs undergoing tank-treading and tumbling motions are a common focus because many numerical methods are capable of simulating isolated cells in shear. Flipping motions are observed at higher viscosity ratios, λ at a low shear rate a reduction in λ can cause a transition from the tumbling to tank-treading regime.^{42–55} An increase in shear modulus serves as a deterrent to tank-treading

motion⁴⁹ and the inclusion of membrane viscosity does not affect the amplitude and inclination angle of RBCs in tank-treading, but allows for better prediction with experiments due to an increase in viscous dissipation.²⁴

The deformation of isolated RBCs in small vessels and capillaries has also been investigated using different numerical methods. Flows through capillaries and divergent geometries as small as 4 μm in diameter have been studied^{25,57} and have demonstrated the large increases in relative viscosity due to complete vascular blockage. Performing simulations in vessels less than the RBC diameter, 8 μm , requires special care for initialization since an initial bi-concave shape cannot be used. Diaz and Barthès-Biesel¹⁸ also demonstrate the capability to simulate the flow capsules in pores and the entrance behavior dependence on internal to suspending fluid viscosity ratio. In slightly larger vessels a characterization of the parachute shape of RBCs has also been investigated.^{24,43–54} The degree of velocity profile blunting is dependent on the rigidity of RBCs and their tendency to migrate toward the vessel core.^{21,56} This deformation induced migration of RBCs has also been studied²² and the equilibrium location is dependent on Reynolds number at low hematocrit levels. Simulations of flow in larger vessels up to 30 μm in diameter have been performed^{20,33,57} to demonstrate the Fåhræus-Lindqvist effect and the change in cell-depleted layer thickness²⁵ with shear rate.

Simulations of Platelet Margination

Details of platelet margination can be modeled mainly due to the increase in scalability of specific computational methods and parallel computing resources. The process of margination requires simulations in extremely long tubes due to the length and time scales in which the process occurs. Two-¹⁴ and three-dimensional⁵⁹ simulations have reported that margination occurs more rapidly at higher shear rates in pressure gradient-driven flow between parallel plates. Hematocrit was also deemed an important factor and it was found that the diffusivity of platelets is dependent on their proximity to the wall. This is primarily due to the change in hematocrit and shear stresses experienced as platelets expel toward the wall. The diffusivity of platelets was predicted to scale linearly with wall shear rates in excess of 2,500 s^{-1} .

METHODOLOGY

Lattice-Boltzmann Method

In order to simulate the process of margination in a rigid tube, a lattice-Boltzmann (LB) based fluid solver is coupled with a deformable RBC membrane model and a rigid dynamics solver for platelets. An extensive

review on the LB method by Chen and Doolen⁸ demonstrates the use of the Chapman–Enskog expansion for recovering solutions to the incompressible Navier–Stokes equations and Aidun and Clausen³ demonstrate its application to several complex flows such as turbulence and particle-laden flows. The fluid solver is based on the D3Q19 single-relaxation-time LB implementation of Aidun *et al.*⁴ and Ding and Aidun¹⁹ and as applied to deformable particles by MacMeccan *et al.*³⁴ and Reasor *et al.*⁴⁵ that was parallelized¹² for simulations of deformable suspensions on distributed memory computing resources.

The Boltzmann equation is discretized in velocity space in the LB method with lattice velocity vectors, $\mathbf{e}_i$ where $i = 1, \dots, Q$, with the value of Q depending on the number of dimensions and the LB stencil chosen. The discrete lattice nodes are denoted by the position vector $\mathbf{r}$. A distribution of fluid particles, f_i , exists at each lattice node for each lattice direction and the time evolution of f_i is governed by streaming and collision operations expressed as

$$f_i(\mathbf{r} + \mathbf{e}_i, t + 1) - f_i(\mathbf{r}, t) = -\frac{1}{\tau} [f_i(\mathbf{r}, t) - f_i^{(0)}(\mathbf{r}, t)], \quad (1)$$

where f_i is the single-particle distribution function, $f_i^{(0)}$ is the equilibrium distribution function, $\mathbf{r}$ and $\mathbf{e}_i$ are the location and velocity of the fictitious fluid particles, τ is the single relaxation time, and t is time. The equilibrium distribution function, based on a Maxwellian distribution, satisfying the Navier–Stokes equations, is

$$f_i^{(0)}(\mathbf{r}, t) = w_i \rho \left[1 + \frac{1}{c_s^2} (\mathbf{e}_i \cdot \mathbf{u}) + \frac{1}{2c_s^4} (\mathbf{e}_i \cdot \mathbf{u})^2 - \frac{1}{2c_s^2} (\mathbf{u} \cdot \mathbf{u}) \right], \quad (2)$$

where ρ is the density, $c_s = \sqrt{1/3}$ is the pseudo-sound-speed, and w_i are lattice weights dependent on the LB stencil used. In the case of the D3Q19 stencil, they are $w_0 = 1/3$ for rest, $w_{1-6} = 1/18$ for the non-diagonal directions, and $w_{7-18} = 1/36$ for the diagonal directions. The kinematic viscosity is related to the relaxation time and the pseudo-sound-speed via $\nu = c_s^2(\tau - 1/2)$. The macroscopic properties are recovered at $\mathbf{r}, t$ via

$$\sum_i f_i^{(0)}(\mathbf{r}, t) = \rho, \quad (3)$$

$$\frac{1}{\rho} \sum_i f_i^{(0)}(\mathbf{r}, t) \mathbf{e}_i = \mathbf{u}, \quad (4)$$

$$\sum_i f_i^{(0)}(\mathbf{r}, t) \mathbf{e}_i \mathbf{e}_i = c_s^2 \rho \mathbf{I} + \rho \mathbf{u} \mathbf{u}, \quad (5)$$

where $\mathbf{I}$ is the identity tensor. In an alternative approach to the Chapman–Enskog expansion, Junk

et al.^{30,31} derive the incompressible Navier–Stokes equations from the discrete LB equation in the diffusive limit using the method of asymptotic expansions.

Fluid-Solid Coupling

The fluid–solid coupling is performed identical to that of the previous work by MacMeccan *et al.*,³⁴ Clausen and Aidun^{10,11} and Clausen *et al.*^{12,13} This method utilizes the method of forming “links” that cross the particle surface that join LB fluid nodes that are interior and exterior to the particle surface. The standard bounce-back (SBB) boundary condition enforces the no-slip condition on RBC, platelet, and vessel wall surfaces through utilizing links to adjust the LB particle distributions on link endpoints. The SBB is second-order accurate in space if the boundary resides at the midpoint between two fluid points⁶⁰ and first-order accurate otherwise.^{26,40} The bounce-back procedure adjusts the distribution on the link endpoints *via*

$$f_i(\mathbf{r}, t + 1) = f_{i'}(\mathbf{r}, t^+) + 6\rho w_i \mathbf{u}_b \cdot \mathbf{e}_i \quad (6)$$

for a link in the i' direction, where i' is the direction opposite of i , t^+ denotes the time post collision but prior to streaming, and $\mathbf{u}_b$ is the local solid velocity at the link intersection point on the surface. Afterward, the traction force on the surface of the particle or vessel wall is calculated and used in the updating of particle dynamics or in the calculation of surface pressure and shear stress on stationary surfaces.

Contact Modeling

This work does not attempt to accurately model the adhesive forces of platelet–platelet or platelet–wall interactions which have been investigated by Mody and King³⁸ since this study is focused on the process of platelet margination and not those due to aggregation or thrombosis growth. The contact modeling used in this work preserves stability and prevents RBC and platelets from overlap as they interact. The functional form of the contact force is

$$A_c \exp\left(\frac{g_c - g}{\sigma_c}\right) \quad \text{if } g \leq g_c, \quad (7)$$

where A_c is a contact scaling constant, σ_c determines the range of the contact force, and g and g_c are the gap distance and the cutoff gap distance respectively. This force is applied normal to the discretized faces at the particle surface then interpolated to the nearest surface nodes prior to updating particle dynamics. The details of the contact model are presented in depth by MacMeccan *et al.*³⁴ and the sensitivity to the specific model parameters is presented by Clausen.⁹

RBC Membrane Model

The RBC membrane model¹⁵ used is rooted in spectrin-link theory and has shown to capture the deformation of RBCs in tension, dynamical regimes of tumbling and tank treading, as well as the deformation of parachuting RBCs in microvessel-sized tubes. The RBC membrane model used is a coarse-grained spectrin-link method based on the research of Pivkin and Karniadakis⁴¹ that was recently shown to be an attractive method to couple with the LB method by Reasor *et al.*⁴⁵

The spectrin-link method used in this work models the Helmholtz free energy of the RBC membrane by considering four contributions

$$E\{\mathbf{x}_n\} = E_{\text{in-plane}} + E_{\text{bending}} + E_{\text{volume}} + E_{\text{area}}, \quad (8)$$

where $\mathbf{x}_n$ is the n th vertex of the RBC membrane triangulation. Each term contributes to the forces that are used to update the dynamics of the n th vertex through

$$\mathbf{f}_n = -\frac{\partial E\{\mathbf{x}_n\}}{\partial \mathbf{x}_n}. \quad (9)$$

The specific forms of each energy term can be found in Pivkin and Karniadakis⁴¹ and Reasor *et al.*⁴⁵ The force terms are found analytically through careful chain-rule differentiation of each term in the Helmholtz free energy equation. The dynamics of each vertex are updated according to Newton's equations of motion via

$$\frac{d\mathbf{x}_n}{dt} = \mathbf{v}_n; \quad M \frac{d\mathbf{v}_n}{dt} = \mathbf{f}_n + \mathbf{f}_n^{\text{LB}} + \mathbf{f}_n^{\text{PP}} = \mathbf{f}_n^{\text{total}}. \quad (10)$$

in which M is taken to be the average amount of mass at each point $M = \frac{1}{N} \int_V \rho dV$ based on the mass of the fluid volume interior to the RBC membrane where V is the volume enclosed by the RBC membrane, ρ is the density of cytoplasm, and N is the number of points used to discretize the membrane surface. Since, the LB method is an explicit first-order accurate fluid solver, a first-order accurate forward Euler scheme is used to update the RBC vertex dynamics. Subgrid modeling is used to prevent the suspending particles from overlap and is given in detail by MacMeccan *et al.*³⁴ and Clausen *et al.*¹³ The forces due to inter-particle forces are denoted by $\mathbf{f}_n^{\text{PP}}$ and those due to the fluid-solid coupling with the LB method are denoted $\mathbf{f}_n^{\text{LB}}$.

Rigid Particle Modeling

The rigid particles used in this work to mimic platelets are updated using a simple Newtonian dynamics solver for rigid oblate spheroids shaped particles with assumed symmetry along their x and y axes. The fluid-solid coupling from the standard bounce-back operation and the particle-particle interactions generates the forces on each triangulated

surface. The forces and torques on each triangle are then summed over the entire particle to get the applied forces and torques.

Setup

For the simulations presented in this article, the flow of suspending fluid and particles is driven by an axial body force used to mimic a constant axial pressure gradient starting from an initially quiescent state. The time scales associated with the margination process are similar to those associated with the periodic nature of the cardiac cycle. However, the Womersley number is small due to the small length scales associated with this problem, i.e., $Wo = R\sqrt{\omega\rho/\mu_a} = \mathcal{O}(10^{-2})$. Therefore, the assumption of a zero Womersley number⁵³ is appropriate. The axial body force magnitude is identical for all the simulations and the flow rate is lower for the higher hematocrit simulations due to the corresponding increase in relative viscosity. The wall shear rates are nearly equal on average, $\dot{\gamma}_w = 355 \text{ s}^{-1}$, due to the presence of the RBC-depleted layer. The reduction in the wall shear rate due to the presence of rigid particles is negligible. The initial orientations and locations of the platelets were seeded randomly, but at a minimum distance of $3 \mu\text{m}$ from the vessel wall. The platelet-shaped particles at nearly the same in all the simulations to mitigate the influence of the initial distribution.

The LB fluid grid resolution used is $3c = 1 \mu\text{m}$ where c is a unit LB spacing with each RBC having a diameter of $24c$ ($8 \mu\text{m}$). Each SL based RBC is constructed of 613 nodes for the surface triangulation with an average coarse-grained spectrin-link length of $0.5 \mu\text{m}$. The resolution of the surface triangulation is sufficient for capturing accurate RBC deformations^{24,41,45} and the coarse-graining procedure of Pivkin *et al.* is necessary for computations containing hundreds of RBCs. Surface triangulations of the resolution used in this work have shown³⁴ to accurately model hydrodynamic forces using the solid-fluid compiling employed. The rigid tubes used in these simulations have a diameter of $41.3 \mu\text{m}$ with a length of $60 \mu\text{m}$, the inlet and outlet boundary conditions are treated as periodic to reduce the computational domain size, and the number of LB fluid grid points used is 2.77 million. The length of the tube should be chosen to mitigate periodicity effects such as clusters of RBCs and platelets spanning the entire length of the tube (see Brady⁶ for details). A detailed list of the simulations performed is given in Table 1. The average velocity, $\bar{u}$, is calculated via $q/\pi R^2$, where q is the volume averaged flow rate and R is the vessel radius. All simulations are computed for 3 million LB time steps which is approximately 6.3 s of physical time. The flow rate (q) for each simulation is listed in Table 1.

TABLE 1. Simulations performed in this study all having a wall shear rate of $\dot{\gamma}_w = 355 \text{ s}^{-1}$.

Sim	ϕ	RBCs	Particles	λ	$\dot{\gamma} \text{ (s}^{-1}\text{)}$	Ca	$q \text{ (mL/min)}$
A	0.2	144	40 Platelets	—	127	Rigid	8.11×10^{-4}
B	0.2	144	40 Platelets	5.0	127	0.097	9.55×10^{-4}
C	0.2	144	40 Platelets	1.0	127	0.097	9.55×10^{-4}
D	0.2	144	40 Platelets	0.5	127	0.097	9.55×10^{-4}
E	0.2	144	40 Disks	5.0	127	0.097	9.55×10^{-4}
F	0.2	144	40 Spheres	5.0	127	0.097	9.55×10^{-4}
G	0.3	216	40 Platelets	5.0	105	0.080	7.64×10^{-4}
H	0.4	288	40 Platelets	5.0	79.8	0.061	5.73×10^{-4}

The cost of the simulations is strongly driven by the number of RBCs and platelets used. The fluid–solid coupling accounts for approximately 50% of the total simulation cost.¹² These simulations were performed on either 64 or 96 computational cores of the Ranger supercomputer located at the Texas Advanced Computing Center. The cost of each simulation ranges from approximately 15,000 core hours to 30,000 core hours depending on hematocrit, ϕ . The total computational cost for this study was approximately 150,000 core hours.

The simulations in this work compare the influence of the shape of rigid platelets with identical volume, viscosity ratio, λ , of the internal to suspending fluid of the RBCs, and hematocrit. A total number of 40 rigid platelets are used and are distributed in random axial locations. The initial radial locations are all chosen within $r/R \leq 0.8$ to ensure every platelet has to travel a minimum distance of $4 \mu\text{m}$ outward in the radial direction to reach the tube wall and are chosen to yield statistical information for all radial locations as the platelets marginate. The initial platelet locations are identical for the three hematocrit simulations. The number of platelets chosen is greater in number (by volume) than observed physiologically for more statistical observations.

The “model” platelet-shaped particles are oblate spheroids and rigid, with a major diameter of $2.3 \mu\text{m}$ and a thickness of $1.0 \mu\text{m}$. Two other shapes are explored; disk-shaped particles with a major diameter of $3.26 \mu\text{m}$ diameter and a thickness of $0.523 \mu\text{m}$; and spherical-shaped particles that have a diameter of $1.73 \mu\text{m}$. These three shapes were chosen to represent limiting cases of non-activated platelets with representative volumes.

The RBC capillary numbers based on the membrane shear modulus defined by $Ca = \mu\dot{\gamma}a/G$, where μ is the plasma viscosity, $\dot{\gamma}$ is the shear rate, a is the RBC radius ($4 \mu\text{m}$), and G is the membrane shear modulus ($6.3 \times 10^{-3} \text{ dynes/cm}$) are also given in Table 1. The capillary number is used to characterize the importance of viscous effects to the elasticity of the RBC

membrane. If the dynamic viscosity of the suspending fluid and the material properties of the RBC are not altered, Ca is a convenient way to present the dependence on shear rate. The shear rates given in Table 1 are based on matching Ca . For all the simulations, the RBC Reynolds number $Re_{\text{RBC}} = \rho\dot{\gamma}a^2/\mu$ is less than 0.6, but larger than physiological conditions. The Péclet numbers for these simulations are very large, therefore Brownian motion is negligible. The motion of platelets and RBCs are driven by hydrodynamic and inter-particle forces and deformation only. Particles smaller than the RBCs (platelet analogues) are treated as rigid particles under the assumption that the particle capillary number is the most important dimensionless parameter. The capillary number for a physiological platelet is much smaller than that of an RBC under similar flow conditions due to their smaller size of $a = 1.15 \mu\text{m}$ and larger shear modulus of $0.03 \pm 0.01 \text{ dynes/cm}$.²⁸

RESULTS

By considering a binary suspension of deformable RBCs and rigid platelets, the influence of hematocrit, viscosity ratio, and platelet aspect ratio are discussed. The experimental results of Aarts *et al.*² and Tilles and Eckstein⁴⁷ demonstrate that the excess of platelets in the near-wall region increases with hematocrit. The results presented herein demonstrate, for a given average axial distance, the near-wall excess of platelets increases with hematocrit. The importance of viscosity ratio is also emphasized. A single simulation with rigid RBCs, representing the extreme viscosity ratio where the interior fluid behaves as a solid, corroborates the experimental findings of Eckstein *et al.*²³ that deformable RBCs are essential to the margination process. Lastly, the aspect ratio of platelets is shown to be significant parameter in the margination process.

Visual Description

Four descriptive images have been constructed, in Fig. 1, to better illustrate the process of margination and the final orientation of a rigid platelet. The RBCs are rendered as transparent to visualize the individual platelet of interest. Figure 1a describes the motion of a typical ellipsoid platelet undergoing margination through highlighting a single platelet in the $\phi = 0.40$ simulation that takes place over $330 \mu\text{m}$. In this case, the highlighted platelet migrates more rapidly than average, but is more applicable for the purpose of visualization. For this particular platelet, the path is also shown from the view looking down the length of the tube in Fig. 1b where the platelet travels in a direct

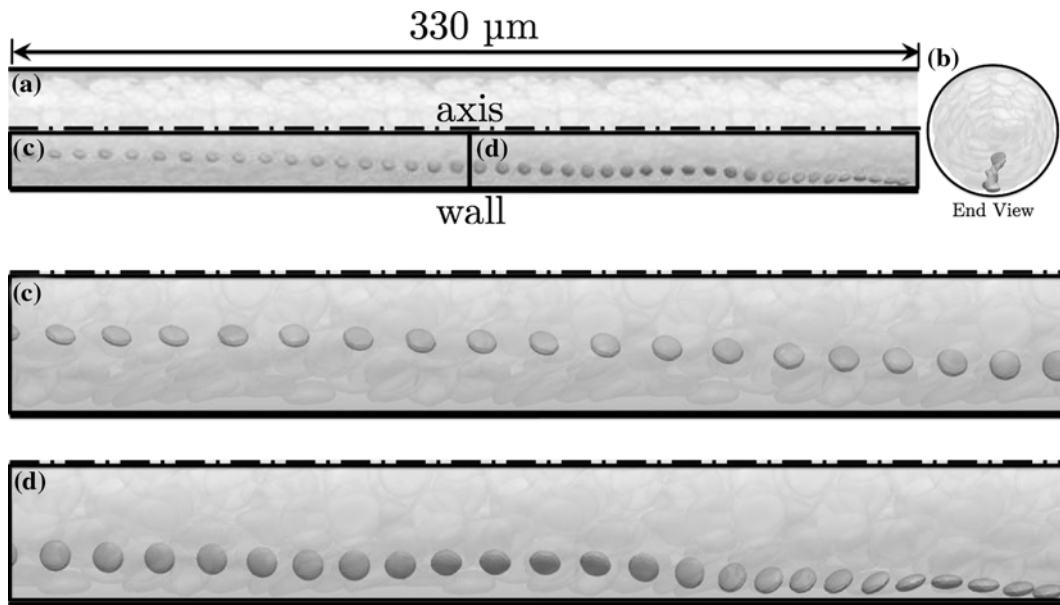


FIGURE 1. The path of an individual platelet that marginates in approximately $330\ \mu\text{m}$ from its initial radial location of $10\ \mu\text{m}$ from the tube wall for the simulation with $\phi = 0.40$ and $\lambda = 5$. Image (a) is a view of the entire length of the tube in which the platelet travels, (b) is the end view of the tube showing the radial path of the platelet, (c) and (d) are detailed views emphasized by the boxes in (a). The centerline and tube walls are indicated by the dashed and solid lines respectively.

path toward the vessel wall. Detailed views of the platelet motion are given in (c) and (d) to show that it rotates and slides along its path to the RBC-depleted layer.

Dependence on Hematocrit, ϕ

In the absence of RBCs, platelets migrate to the Segré–Silberberg equilibrium position⁴⁶ of $r/R \approx 0.6$. Using the method presented in the previous section, platelets demonstrate rapid migration to the S–S equilibrium position (as seen in Fig. 2) demonstrating the tubular-pinch effect which is observed in the experimental studies of Aarts *et al.*,² Uijttewaal *et al.*,⁵² and Nijhof *et al.*³⁹ The phenomenon of margination is complicated by additional forces from RBC interactions. One commonly studied parameters in the margination process is hematocrit, ϕ . A larger percentage of the platelets will accumulate at the vessel wall with increasing hematocrit. The RBC-depleted layer thickness is a strong function of the hematocrit as the RBCs reach a dense packing fraction along the tube axis. In Fig. 3a the average platelet trajectories for the $\phi = 0.20$, 0.30 , and 0.40 simulations are given for a physiological viscosity ratio of $\lambda = 5$. In this figure, δ denotes the distance to the vessel wall in microns and $\int_0^T \bar{u}_x dt$ denotes the average distance traveled in millimeters. In these simulations, the axial body force used is the same which results in identical wall shear rates, but a reduction in flow rate as the hematocrit is increased. The effect of increasing the hematocrit is apparent as the rate of

margination is increased with increasing hematocrit which is emphasized in the figure. For the physiological hematocrit of $\phi = 0.40$, nearly all the platelets have migrated to the vessel wall after traveling only $3\ \text{mm}$. For $\phi = 0.20$, this distance is greater than $8\ \text{mm}$. The difference in slope of these results highlights the increase in margination rate for the same axial pressure gradient. The relative viscosities ($\mu_r = \mu_a/\mu_{\text{plasma}}$) of the simulations with $\phi = 0.20$, 0.30 , and 0.40 are 1.40 , 1.70 , and 2.23 respectively.

Peak concentrations of platelets in the near-wall region from these simulations compare favorably to the experiments of Aarts *et al.*² as seen in Table 2. Figure 4 compares the simulation results for $\phi = 0.40$ with the experimental studies of Aarts *et al.* displaying the peak location in platelets near the wall. The tube used in the studies of Aarts *et al.* is $3\ \text{mm}$ in diameter, which is considerably larger than the tube size used in the present simulations of $41.33\ \mu\text{m}$. This difference explains the difference in peak location as a function of r/R . Zhao *et al.*⁵⁸ show that the peak concentrations presented by Aarts *et al.* appear to be coincident with the wall due to insufficient resolution in their measurements and that the peak should be at approximately $2\ \mu\text{m}$ from the wall.

Figure 5 demonstrates the differences between the three simulations with varying hematocrit, ϕ . The initial platelet distributions along with the evolution of the distributions are given at several downstream locations. The overall result is consistent with experimental observations which show that the near-wall

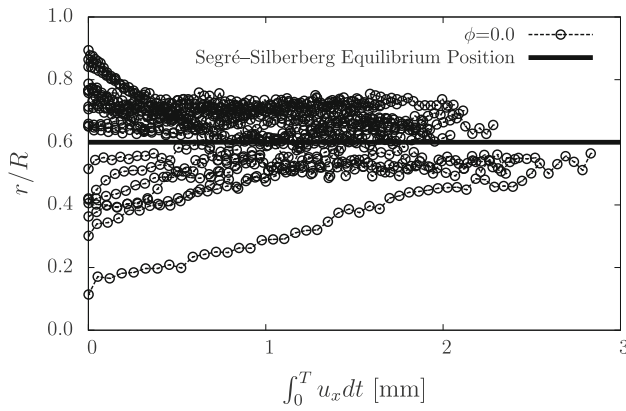


FIGURE 2. Platelet-shaped particle trajectories for $\phi = 0.0$ demonstrating the migration to the Segré-Silberberg equilibrium position of $r/R \approx 0.6$.

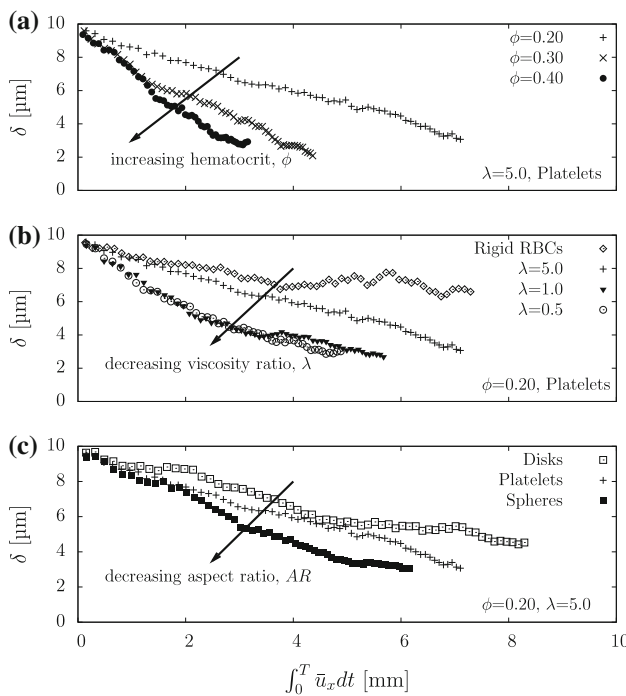


FIGURE 3. Average rigid particle trajectories for all simulations: (a) 40 platelets with $\phi = 0.20$ (+), 0.30 (x), 0.40 (●) and $\lambda = 5.0$; (b) 40 platelets with $\phi = 0.20$, $\lambda = 5.0$ (+), 1.0 (▼), 0.5 (○) and rigid RBCs (◇); (c) 40 spheres (■), platelets (+), and disks (□) with $\phi = 0.20$, $\lambda = 5.0$. The distance to the microvessel wall δ , given in microns, is plotted against the distance traveled in the flow direction in millimeters. Arrows indicate trends for increased margination.

excess of platelet shaped particles increases with increasing hematocrit. The peak location is nearer to the wall for the realistic hematocrit values of $\phi = 0.30$ and 0.40 due to the orientation of the platelets in the cell-depleted layer and the cell-depleted layer thickness. For the $\phi = 0.30$ and 0.40 cases, the platelet-shaped particles slide along the vessel surface for the majority of the time once they reach the cell-depleted layer. For

TABLE 2. Comparison of platelets distributions (%) in the near-wall region for hematocrit values of $\phi = 0.20$ (B), 0.30 (G), and 0.40 (H) for platelet shaped particle simulations.

Source	Wall shear rate, $\dot{\gamma}_w$ (s^{-1})	Hematocrit, ϕ		
		0.20 (%)	0.30 (%)	0.40 (%)
Experiments	240	59	—	72
Simulations	355	60	62	77
Experiments	760	66	—	79

Experimental results are provided from Aarts *et al.*¹⁵

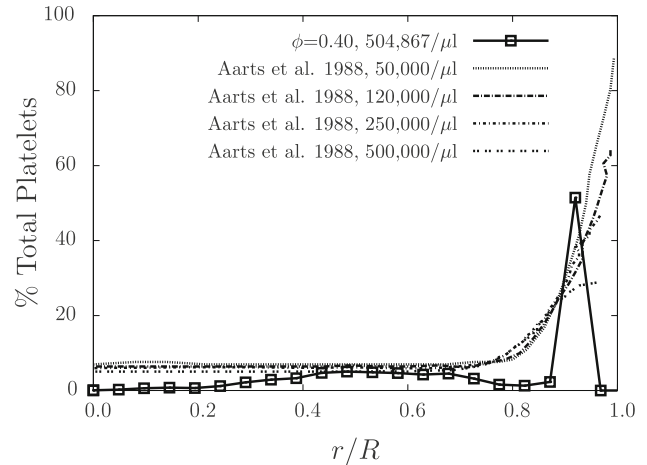


FIGURE 4. Comparison of platelet concentrations with the experimental results of Aarts *et al.*² The experiments were performed with four levels of platelet concentrations and wall shear rate of $\dot{\gamma} = 1260 s^{-1}$ in a 3 mm diameter tube and $\phi = 0.40$.

the reduced hematocrit simulation with $\phi = 0.20$, the particles undergo flipping motions in the cell-depleted layer. The platelet rotation rates may be influenced by adhesion in the vicinity of the wall, but this effect is not quantified in this study. The authors refer the reader to the study of Mody and King³⁸ for a comprehensive study on the effects of inter-particle forces in close proximity and far from walls.

Dependence on Viscosity Ratio, λ

For $\phi = 0.20$, numerical experiments for $\lambda = 0.5, 1, 5$ and with completely rigid RBCs have been performed. The average trajectories of platelet shaped particles for the values of λ simulated are given in Fig. 3b. This figure displays that λ is an important parameter influencing the rate of margination. The lower viscosity ratios correspond to an increase in the margination rate as emphasized in the figure. In the rigid RBC case, the RBC-depleted layer does not develop near the vessel wall and platelets not initially in the vicinity, i.e., $\delta \leq 5 \mu m$, oscillate randomly with no indication of margination.

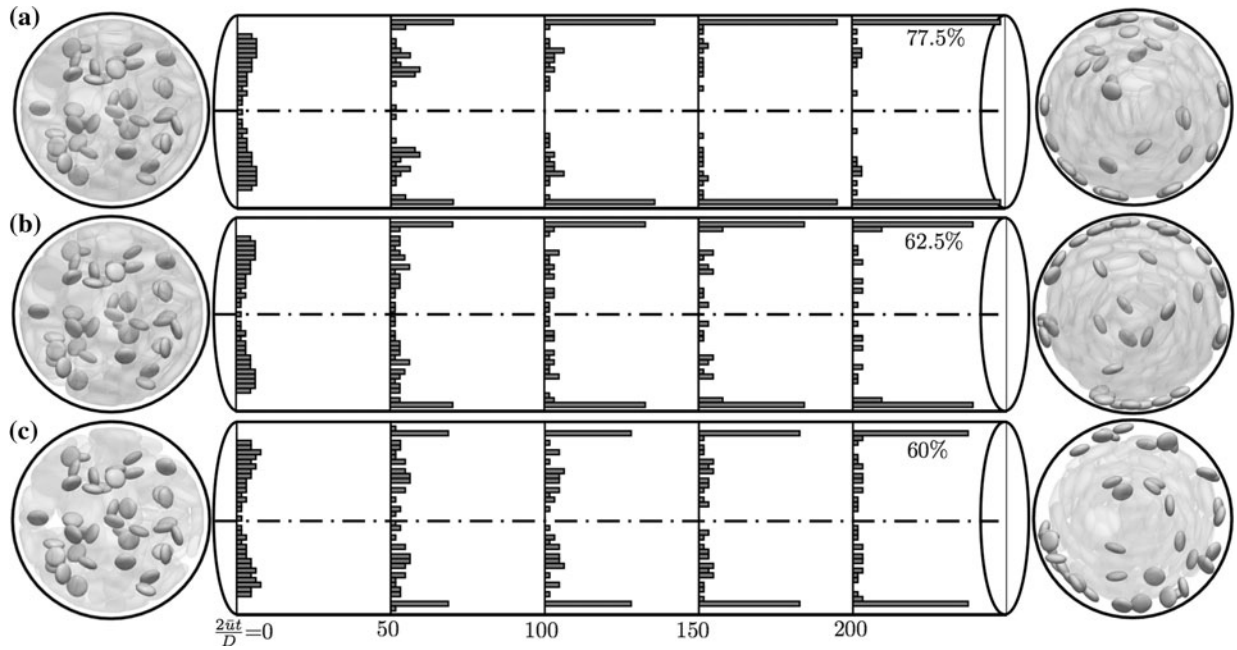


FIGURE 5. Temporal evolution of platelet distributions for (a) $\phi = 0.40$, (b) $\phi = 0.30$, and (c) $\phi = 0.20$ for platelet shaped particles with $\lambda = 5$. At $2\bar{u}t/D = 200$ the platelet concentrations are labeled. The bin width for the concentration profiles is $w = 1 \mu\text{m}$. The initial and ending end views of the tube are also given to display the distribution of platelets before and after margination.

The rigid RBCs are restricted to the tumbling motion only and do not migrate toward the vessel center. Therefore, computing the simulation for any longer amount of time was deemed unnecessary.

The effect of viscosity ratio, λ , is demonstrated in Fig. 6 through visualizing the evolution of radial platelet distributions. The evolution of the platelet distribution for the rigid RBC case is notably different than the other three simulations. Rigid RBCs do not migrate to the vessel core and do not undergo tank-treading motions in the high shear area near the wall. With rigid RBCs platelets do not exhibit the rapid migration toward the tube wall and the lack of a cell-depleted layer prevents them from remaining in the area of high shear stress. The cell-depleted layer thickness is similar for the cases with $\lambda = 0.5, 1$, and 5.0 , but the cases with $\lambda \leq 1$ reveal larger platelet peak concentrations near the wall with a more rapid evolution. This observation is consistent with the experiments of Zhao *et al.*⁵⁸ performed with suspending fluid viscosities greater than blood plasma.

Dependence on Platelet Aspect Ratio, AR

For all the shape study simulations performed, the hematocrit is $\phi = 0.20$ and the initial locations of the rigid particles and RBCs are nearly equal, but differ due to the difference in their size and proximity to the deformable RBCs. In Fig. 3c the average trajectories of the three simulations are given using the

physiological viscosity ratio of $\lambda = 5$. The aspect ratio is denoted as AR in the figure where spheres have an aspect ratio of $AR_{\text{sphere}} = 1$, $AR_{\text{platelet}} = 2.3$, and $AR_{\text{disk}} = 6.24$. This plot demonstrates that the different shapes have unique average margination rates, but confirm that all the shapes investigated experience margination. The tendency of the disk shaped particles to reach an final position against the RBCs on the edge of the RBC-depleted layer (emphasized in Fig. 7c) results in a larger final δ value than the spherical or ellipsoid platelet-shaped particles. The spherical-shaped particles migrate the most rapidly and the majority reach the outer layer after traveling an average distance of 6 mm .

The effect of platelet aspect ratio is further demonstrated in Fig. 7. The RBC-depleted layer thickness for these three simulations are similar, but the orientation of the platelets differ depending on aspect ratio. Assuming a log-normal distribution and keeping the data within two standard deviations, the mean distances to the wall at time $2\bar{u}t/D = 200$ are $\bar{\delta}_{\text{sphere}} = 1.41 \mu\text{m}$, $\bar{\delta}_{\text{platelet}} = 1.69 \mu\text{m}$, and $\bar{\delta}_{\text{disk}} = 2.24 \mu\text{m}$ with standard deviations of $\sigma_{\text{sphere}} = 0.057 \mu\text{m}$, $\sigma_{\text{platelet}} = 0.161 \mu\text{m}$, and $\sigma_{\text{disk}} = 0.311 \mu\text{m}$. Therefore, the mean distance to the wall and standard deviation both decrease with decreasing aspect ratio. The spherical and model platelet-shaped particles interact simultaneously with the vessel wall and the RBCs at the edge of the cell-depleted layer. The thin disk-shaped particles align adjacent to the RBCs on the edge of cell-depleted layer and do not interact with the

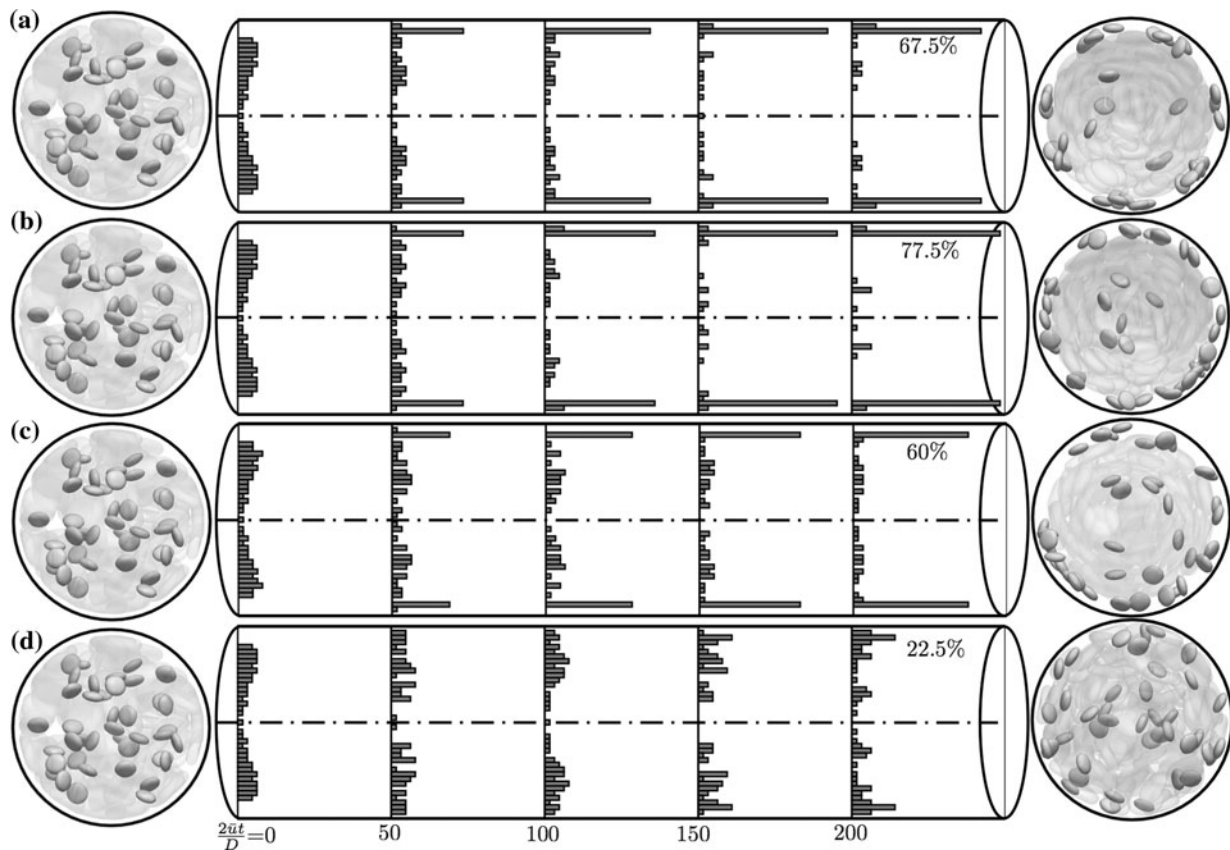


FIGURE 6. Temporal evolution of platelet distributions for $\phi = 0.20$ for (a) $\lambda = 0.5$, (b) $\lambda = 1$, (c) $\lambda = 5$, and (d) rigid RBCs with platelet shaped particles. At $2\bar{u}t/D = 200$ the platelet concentrations are labeled. The bin width for the concentration profiles is $w = 1 \mu\text{m}$. The end views of the tube are also given to display the distribution of platelets before, $t = 0$, and after margination, $t = 100D/\bar{u}$.

vessel wall unless they undergo a flipping motion. The peak concentrations at the wall are larger for the spherical particles and the distributions evolve more rapidly than the disk-shaped particles. The high aspect ratio of the disk-shaped particles allows them to slide between RBCs more easily than the lower aspect ratio particles.

DISCUSSION

Using a hybrid lattice-Boltzmann spectrin-link suspension cellular blood flow solver⁴⁵ the process of margination has been investigated for flow through a rigid $41.3 \mu\text{m}$ microvessel. Our simulations indicate that the process of margination develops fairly rapidly ($< 1 \text{ cm}$) for all the deformable RBC configurations tested. The influence of hematocrit and viscosity ratio, both have been demonstrated experimentally, are captured using the LB-SL method with rigid platelets. Through choosing limiting cases for the platelet morphology, the results indicate that it is also an important parameter that can alter the rate of margination. The

effect of shear rate, $\dot{\gamma}$ (or Ca_G) on platelet margination has been documented through experimental^{47,58} and computational⁵⁹ studies and is therefore not considered in this work.

Platelets at identical initial locations marginate more rapidly under physiological hematocrit with $\phi = 0.40$, rather than 0.20 , which is consistent with experimental observations.² At equal axial pressure gradients, the RBC-depleted layer thickness is smaller for $\phi = 0.40$ than $\phi = 0.30$ or 0.20 . This reduction in RBC-depleted layer thickness for $\phi = 0.40$ results in the platelet-shaped particles aligning their minor axis with the radial coordinate and exposing more surface area in the high shear region adjacent to the wall. Platelet-shaped particles undergo flipping motions as they interact with the wall and the RBCs on the edge of the RBC-depleted layer for the smallest hematocrit level simulated ($\phi = 0.20$), which is not observed in the $\phi = 0.40$ simulations. The dominance of sliding motion at physiologically relevant hematocrit levels is consistent with the intravital studies of Maxwell *et al.*³⁵ performed at $30\text{--}40 \mu\text{m}$ rat arterioles where the majority of platelets were observed to slide at

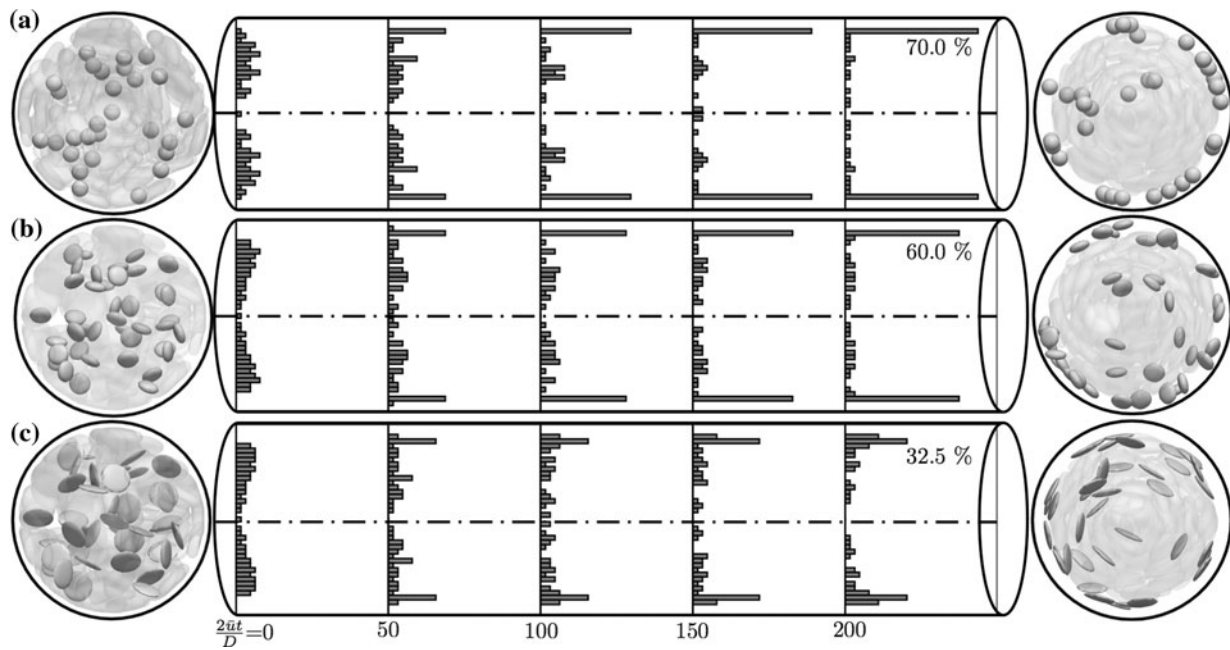


FIGURE 7. Temporal evolution of platelet distributions for $\phi = 0.20$ for (a) spheres, (b) platelets, and (c) disk shaped particles with $\lambda = 5$. At $2ut/D = 200$ the platelet concentrations are labeled. The bin width for the concentration profiles is $w = 1 \mu\text{m}$. The initial and ending end views of the tube are also given to display the distribution of platelets before and after margination.

thrombotic injury sites or vessel walls rather than going through flipping motions. The sliding motion of platelets in the RBC-depleted layer has important consequences in kinetics of platelet adhesion to injured vessel walls by maximizing the contact area between the platelet surface and the adhesion site compared to the flipping motion. However, the accuracy of the sliding dynamics in these simulations has not been evaluated.

Spherical particles were shown to migrate more rapidly than higher aspect platelets with identical volumes. The ellipsoid-shaped particles and thin disks marginate less rapidly due to their tendency to flip and slide between RBCs. This ability to slide between RBCs may hinder the rate of margination by trapping them between RBCs. This is emphasized by the disk shaped particles marginating at the slowest rate. If the largest dimension of a model platelet is larger than the RBC-depleted layer thickness, the orientation of the particle near the vessel wall will be different than the case when it is smaller than the RBC-depleted layer thickness. The hematocrit is reduced, such that the RBC-depleted layer thickness is larger than $2.5 \mu\text{m}$ for model platelet simulations. Then the orientation of the platelet in the RBC-depleted layer can drastically affect the shear stress history on the platelet and its influence on platelet activation. Accurately capturing the shear stress history on platelets interacting with the endothelial wall requires the addition of a representative

adhesion model capable of capturing the forces and torques measured in experiments.

The ratio of the viscosity of the interior fluid of the RBC membrane to that of the suspending fluid, λ , is an important dimensionless parameter in RBC dynamics. As λ decreases, the shear rate threshold where tank-treading behavior is observed also decreases. Therefore, by decreasing λ there is an increase in the number of RBCs undergoing tank treading motion.⁵⁵ The experimental observations of Zhao *et al.*⁵⁸ demonstrated that the near-wall concentration of platelets is elevated for $\lambda = 0.38$ compared to experiments with $\lambda = 2.5$ in pressure-gradient driven flow in a rectangular channel. Therefore, the rate of margination in experiments where $\lambda \leq 1$ is likely to be larger than for an identical shear rate *in vivo*. Since the rate of margination is increased by decreasing λ to values ≤ 1 , tank treading may be a mechanism for enhancing margination.

Based on the results obtained for the $41.3 \mu\text{m}$ diameter vessel simulations, and assuming the rate of margination is similar in large vessels, the required distance for the majority of platelets to migrate to the vessel wall can be estimated. Assuming the rate of margination is consistent for a wall shear rate of approximately 350 s^{-1} , the estimated distance required for the majority of platelets to marginate in a 1 mm diameter tube with $\phi = 0.40$ would be approximately 100 mm. Based on the margination rates, lower

hematocrit values would yield longer estimates while transitioning to more spherical-like platelets and lower viscosity ratios would yield shorter distance estimates. The lack of an adhesion model likely does not influence the rate of margination, but could alter the mechanics of the platelet shaped particles interacting with the vessel wall. Furthermore, none of the particles demonstrated a tendency to travel back to the core of the vessel after reaching the cell-depleted layer.

The total cost of all the simulations was approximately 150,000 core hours. Doubling the vessel size will increase the computational cost by a factor of eight with an equal L/D . The LB-SL implementation is capable of scaling to $\mathcal{O}(100,000)$ particles, but the computational cost associated with such simulations prevents the study of a large parameter space. Therefore, future simulations with a larger number of particles will focus on comparing to a specific experimental study with alterations to a single parameter such as hematocrit, viscosity ratio, or flow rate.

Platelet-rich thrombi can occlude atherosclerotic arteries where such occlusive thrombi can cause rapid ischemic cardiac arrest. The importance of various parameters, such as those investigated here, on transport of platelets to the wall are important factors in thrombus growth rate. Better insight in the parameters influencing platelet margination will guide us in identifying risk factors associated with atherothrombosis. Furthermore, understanding the influence of physiological factors influencing platelet margination may shed light on ways to prevent arterial occlusions.

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